

# TYLER NORCROSS

(413) 663-4963 | tylernorcross04@gmail.com | linkedin.com/in/tyler-norcross | https://github.com/NightGhost4 | https://tylernorcross.com

## EDUCATION

University of New Hampshire  
B.A. Computer Science, Algorithms

Durham, NH  
May 2026

## PROFESSIONAL EXPERIENCE

### Agent Integrator

Founding Engineer

Remote

December 2025 – May 2026

- Served as **founding engineer** at an AI-native venture studio that embeds inside businesses to design and ship **agentic systems & orchestration layers**; oversaw architecture for **10+ client-facing** software products spanning finance, health-tech, finance, and operation automations.
- Built **full-stack** internal operations platforms and public-facing web applications across multiple simultaneous client engagements; include **API integrations, headless CMS portals, content generation workflows, chatbot tools**, and more.
- Worked across a **modern tech stack** including Next.js, Node.js, TypeScript, PostgreSQL, Claude APIs, and third-party data APIs; contributed to 10+ different system architectures, visual design, production-grade code, **prompt engineering** for agentic workflows, and deployments.

### TaskMind

Technical Co-Founder

Durham, NH

May 2025 – January 2026

- Co-founded an **AI-powered market intelligence SaaS platform**; led **full-stack architecture** using Next.js, Node.js/Express, TypeScript, **Google Cloud Platform (GCP)** via App Engine, MongoDB; designed **50+ RESTful API endpoints** with **multi-tenant data isolation** and **JWT-based role access control**; implemented **end-to-end Stripe subscription infrastructure** with **tiered pricing** and **usage-based metering**.
- Engineered a **multi-model AI orchestration layer** and a **RAG enrichment pipeline** integrating a **dual-sanitization research system** using **Perplexity AI**; features included **vertical reporting, deep company research, competitive insights**, and **marketing campaigns**.
- Awarded a place in the **Berkshire Innovation Center Stage 2 Accelerator (S2A)**; a competitive, equity-free 35-week program awarding a small cohort of early-stage tech startups a dedicated workspace, lab access, and a curated network of industry experts in Pittsfield, MA.

### Dialogix

Full Stack Developer

Research Conference

April 2026

- Built a **full-stack AI conversation coaching platform** in Node.js/Express and **React Native (Expo)** with **JWT + bcrypt authentication, per-user request throttling**, and a **MongoDB schema** tracking conversation state with **idempotent transitions** to prevent data corruption on retries; implemented a **psychometric personality profiling engine** that infers defined core competencies.
- Engineered a RAG pipeline fusing **Pinecone vector search (OpenAI text-embedding)** with **proprietary data** from a sponsor-sanctioned Learning Management System (**LMS**); included a **three-framework leadership model** to generate **context-aware coaching suggestions**.
- Awarded **1st place** in the **UNH Undergraduate Research Conference** (computer science department); this annual celebration of academic excellence at UNH is one of the largest conferences of its kind in the country.

## PROJECTS

### RoomNet

Roommate Matching Platform for University Housing

University Startup Competition

April 2025

- Built a **full-stack AI conversation coaching platform** in Node.js/Express and **React Native (Expo)** with **JWT + bcrypt authentication, per-user request throttling**, and a **MongoDB schema** tracking conversation state with **idempotent transitions** to prevent data corruption on retries; implemented a **psychometric personality profiling engine** that infers defined core competencies.
- Engineered a RAG pipeline fusing **Pinecone vector search (OpenAI text-embedding)** with **proprietary data** from a sponsor-sanctioned Learning Management System (**LMS**); included a **three-framework leadership model** to generate **context-aware coaching suggestions**.
- Placed in **Semifinals** for the **UNH Holloway Competition**; a premier business plan competition for undergraduate and graduate students. With over \$40,000 in prizes, it challenges students to develop products and present their plans to bring them to market.

### Visual Design Agent

UI/UX Development Tool

Design Tool

April 2026 – Present

- Architected a **multimodal AI feedback loop** in Python that generates React/Tailwind components via **Claude Haiku** and **hero images** with **OpenAI GPT Image 2** and renders them at **1440×900** and **390×844** via **headless Chromium (Playwright)**, and pipes screenshot bytes to **Claude Sonnet** for **structured JSON visual critique**; averaging **4 iterations** to reach a **reference quality threshold**.
- Designed a **5-axis scoring system** with **precision fix instructions** constrained to not regress high-scoring axes; deployed on **Hetzner VPS (Ubuntu)** with a **4-stage test suite** validating each pipeline layer; every basis of visual reference is derived from **professional templates**.

## TECHNICAL SKILLS

**Languages & Frameworks:** TypeScript, JavaScript, Python, Java, C, SQL, React, Next.js, Node.js/Express, React Native/Expo, HTML/CSS

**AI & Frontend:** OpenAI API, Claude API, Perplexity API, RAG Pipelines, Prompt Engineering, LangChain, Pinecone, Tailwind CSS, Vite

**Databases & Cloud:** MongoDB, PostgreSQL, MySQL, SQLite, Firebase, Redis, GCP, Docker, CI/CD, GitHub Actions

**Tools & APIs:** REST, Stripe, Git/GitHub, Playwright, Prisma, Postman, Agile/Scrum, Linux/Unix, VPS, SSH